

In-Situ SOP Materials Inventory

This inventory covers 41 of the 44 SOPs currently listed in In_Situ_SOPs/README.md. Load and Run the Oven Tray, Check Out Tools from the Crib, and Return Tools to the Shadow Board are listed in that README but are not yet itemised here. Quantities are per episode unless noted otherwise. Fixed installations are separated from movable tools and consumable or input materials so the list can be used for procurement and station setup.

Materials by task

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Feed a Conveyor Line Station	Stub conveyor with running belt; evenly spaced white pitch marks; near and far rails; overhead light bar and conduit; entry and transfer flags; takt lamp; read-only count panel; fixed load shelf; orientation nest with rib	Parts tote; reject bin	10 square-based housing parts , plus 1 misfeed part pre-positioned on the belt (11 parts total if the misfeed is additional to the tote quantity)
Fit a Fitted Sheet to the Mattress	Bed frame with a cot-size mattress lying square and firm on it; no rail, headboard, or footboard rising above the mattress top; clear access to the top face, all four side faces, and the bottom edge; clear floor along the near long side for the base	None beyond the installed bed and mattress	1 cot-size fitted sheet cut to fit the mattress, with unbroken elastic hem all round, staged spread on the mattress pockets down and unhooked
Grease a Machine Round	Stopped and isolated machine; lockout tag; 3 same-size grease zerks; guard shelf; frame rail; hose clip; tool shelf; fixed grease-gun cradle; card holder	Primed lever grease gun with hose and fitting coupler; blue wipe rag; red purge rag; grease pencil	Grease sufficient for 9 full strokes (3 strokes at each of 3 zerks); lube card with at least 1 empty three-box row
Load the Dishwasher	Installed dishwasher with working latch, control panel, upper and lower racks, and clipped cutlery basket; draining board; utensil cup	None beyond the installed racks, basket, draining board, and utensil cup	2 plates, 2 bowls, 2 cups, 2 forks, 2 knives, 2 spoons

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Mop a Floor Zone	Defined hard-floor bay with back kick rail and left/right plinth ends; fixed mop rest	Rectangular bucket with wringer and fill line; handled water jug; flat mop with fitted absorbent pad; folding A-frame caution sign	Clean water: 1 jug , measured to the bucket fill line; 1 clean, dry mop pad fitted to the mop
Refill the Kettle and Dispenser at the Sink	Sink with working cold-water mixer tap; counter; kettle power base; fixed dispenser with open-top filler tank, front fill line, own tap, and drip tray	Electric kettle; separate fill jug with handle, pouring lip, and fill line; absorbent cloth	Cold tap water sufficient to fill the partly filled kettle to its line and to fill the jug once so the partly filled dispenser tank reaches its line
Organize the Freezer In Situ	Installed freezer with open left-hinged door on a working door catch; working shelf with a shelf directly above; back wall; clear floor space in front for the work stand	Work stand; flat work tray with a bag spot and a box spot; 2 open-topped bins with front rails, one labeled BAGS in blue and one BOXES in green	6 dated items: 3 sealed freezer bags (dates 1, 2, 3) and 3 rigid boxes (dates 1, 2, 3), each with a readable date tag and each bag with a blank label patch; 1 label sheet carrying exactly 2 labels , naming bag 2 and bag 3
Restock the Pantry Shelves	Pantry shelf unit with working shelf and shelf directly above; three zone labels (CARTONS, JARS, CANS) fixed to the shelf front edge; clear floor space in front for the stand	Stand; plain cardboard delivery box with no top flaps and untaped, untucked bottom flaps (collapsible)	Older stock already in place: 1 carton, 1 jar, 1 can. New stock: 2 cartons, 2 jars, 2 cans (6 new items); 1 delivery box flattened per episode
Restock the Refrigerator In Situ	Installed refrigerator with open left-hinged door; working shelf; crisper drawer; tall and short door bins; clear adjacent counter	Stock tote	Older stock already in place: 1 carton, 1 jar, 1 tub, 1 produce pack, 1 bottle, 1 small jar. New stock: 1 carton, 1 jar, 1 tub, 2 produce packs, 1 bottle, 1 small jar (7 new items)

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Run a Microwave Heating Cycle	Installed microwave with turntable, front controls, working door/latch, and clear heat-proof counter	Heat-proof plate with broad rim; vented dome cover with tab; food tub; spoon	Thick prepared food sufficient for 3 level spoonfuls , with some food remaining in the tub
Squeegee a Mirror Panel	Fixed framed mirror panel; clear counter with bottle, cloth, and squeegee positions	Trigger spray bottle set to fan spray; straight rubber-blade squeegee; absorbent lint-free cloth	Glass cleaner sufficient for at least 6 trigger bursts
Stage Outbound Lanes	Fixed dock deck with a smooth top and an empty upper shelf directly above its lane strip; 2 route lanes taped as 3 numbered spots each (Lane A from the left end, Lane B from the right end); ROUTE A and ROUTE B signs on the front edge of the upper shelf; 2 hinged lane flags with a catch (OPEN flat / STAGED standing) at the head end of each lane; fixed manifest card holder at the middle of the upper shelf edge	None beyond the installed lanes, flags, and holder	6 same-size closed cases (shoebox size) with front-face route labels A1, A2, A3, B1, B2, B3, reused; A3 and B1 weighted and carrying a red HEAVY band; 1 manifest card (ROUTE A: 3 cases, ROUTE B: 3 cases, heavy first, then by stop), reused
Tend a 3D Printer	Installed 3D printer with magnetic bed, locating pins, parked toolhead, and working control panel; plate rest/work area	Flexible steel build plate; debris tray; stiff plastic scraper; soft brush	1 completed printed part attached to the plate, with skirt and purge line; 1 next print job ready in the printer queue
Tidy a Bookshelf	Bookcase with a bare working shelf, shelf above, fixed end panels at both ends, and a lower shelf holding the books and non-books; small side table beside the bookcase holding the bin and bookend	Open bin; damp cleaning cloth (ends on the side table); solid freestanding bookend	6 books , all visibly different heights; 3 non-book objects : 1 mug, 1 small picture frame, and 1 coiled cable; visible removable dust on the shelf lip

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Unload the Dishwasher to the Cabinet	Installed dishwasher with working upper and lower racks and clipped cutlery basket, door open flat; open-fronted two-shelf cabinet standing on the counter immediately right of the machine, with 2 up-pointing mug hooks fixed under the front edge of the upper shelf; counter drawer directly below the cabinet, drawn open past the counter edge	Three-compartment cutlery drawer insert (forks, knives, spoons)	2 plates, 2 bowls, 2 mugs with handles, 2 forks, 2 knives, 2 spoons , all clean and dry, staged in the racks; no consumables
Unload the Dryer	Installed front-loading dryer on a pedestal or stack with the drum mouth about level with the arms, left-hinged door that stays open on its own and latches when pressed, pull-up lint filter with a tab in a slot at the bottom of the door opening; low table in front of the dryer below the door opening; hamper stand to the right of the drum mouth, level with the table and clear of the door swing	Small light open-top hamper with one stiff hoop carry handle across the top; open lint bin	6 garments: 3 cotton t-shirts and 3 hand towels , dry, loose on the drum floor; 1 lint sheet per episode pressed onto the filter mesh
Load the Washing Machine	Installed front-loading washing machine on a pedestal with the drum mouth about level with the arms, left-hinged door, almost closed but not latched at the start, that stays open on its own and latches when pressed, clicking program dial, Start button, water tap shut off; low table in front of the machine below the door opening	Low open-top hamper (rim below the open door's bottom edge); small pocket dish	6 garments: 2 cotton t-shirts, 2 hand towels, 2 pairs of cotton shorts with two open front pockets , dry, reused; 1 folded tissue in one pocket per episode

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Tend a CNC Mock Cell	Installed CNC milling mock on its own stand, powered, cuts nothing, spindle parked high at the back; two sliding door leaves meeting in the middle with upright handles; vise bolted to the table behind the door opening, fixed jaw at the back with a scratched middle mark, ledged jaws, handle with a knob and a click-out hub; chip trough behind the vise; front ledge below the door opening; control panel on the right with Cycle Start, Feed Hold, emergency stop, DOOR lamp, and CYCLE lamp	Stock tray; finished tray; rubber bulb blower with a stiff nozzle and its blower cup	1 aluminium stock block (about 60 x 40 x 30 mm, plain) per episode, reused; 1 finished part (same block with a pocket cut in its top) clamped in the vise per episode, reused; about 20 light plastic chips scattered over the vise per episode
Handheld Vacuum Pass and Dock	Fixed raised platform in front of the base at about work-table height; small flat short-pile rug with a bound edge and non-slip backing lying on it, left binding at the middle of the base; fixed dock cradle with a spring clip at the platform's right end, holding the vacuum level	Cordless handheld vacuum with a pistol handle, latching power button and power light on top, clear dust cup with an underside flap and a left-side release catch, fixed wide flat nozzle; open-top bin on the platform's left	1 light, even scatter of dry crumbs over the whole rug per episode; vacuum charged enough for three lanes and one edge pass at full power
Put Away to Shelving Locations	Fixed two-level shelving unit with a shelf above each level and a middle upright; level A: 4 fixed open-front bins (A1 to A4); level B: 2 marked case lanes with back stops (B1, B2); barcoded location label under each location; scanner holster and open-top list box on the middle upright	Auto-read barcode scanner linked to the put list (green light and beep / red light and buzz); two-deck put-away cart with a list clip on the handle, braked; 1 open put tote	2 closed shoebox-size cases and 4 small boxed eaches , each with an SKU label, reused; 1 put list card per episode (CASES then EACHES, line order shuffled)

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Batch-Pick to a Cart	Fixed two-bay shelving (Bay 1 left, Bay 2 right) with a shelf above each of 2 levels, 8 fixed open-front bins (1-A1 to 2-B2) with bin labels; raised track deck in front, below the bottom bins, with a straight two-rail track, an end block at each end, stop lines (left end, STOP 1, STOP 2, right end), and swappable START and PACK signs	Short low pick cart on the track with an upright handle and a brake lever at each end, a four-slot divided tray (A to D), a fixed cart scanner on a post (green/red), and a pick-list clip	24 small boxed units , 3 each of 8 SKUs , reused; 1 pick list per episode (three lines per bay, printed in the config's walk order)
Case-Pick to a Pallet	Fixed three-level rack (A top, B middle, C bottom) with a beam or deck above each level and a middle upright; 2 positions per level (A1 to C2), position label under each; fixed auto-read scanner holster on the middle upright between levels B and C	Auto-read barcode scanner linked to the pick ticket (green light and beep / red light and buzz), stays holstered; small mock pallet (three cases wide, one case deep) on a braked fixed-height pallet stand with a ticket clip	6 same-size closed cartons , about twice as long as wide, SKU label on one end, one-gripper light, reused; 1 pick ticket card per episode (six lines, shuffled)
Pick an Eight-Line Order	Two fixed shelf bays side by side (Bay 1 left, Bay 2 right) with a middle upright and a shelf above each level; 2 levels per bay, 2 fixed open-front bins per level (1-A1 to 2-B2, 8 bins); location label under each bin; fixed auto-read scanner holster on the middle upright	Auto-read barcode scanner linked to the pick list (green light and beep / red light and buzz), stays holstered; flat-top pick cart with list clip on the handle, taped fill spot and STAGED spot with a stop lip, braked; 1 open order tote with a clear list pocket	24 small boxed units , 3 each of 8 SKUs , SKU-labelled, reused; 1 pick list card per episode (BAY 1 then BAY 2, line order shuffled within each bay)

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Hang Parts in a Spray Rack Mock	Raised deck with a straight two-rail cart track, taped load mark at its left part; fixed three-walled booth mock at the right end (open front and left side) with a booth stop bar across the track	Two-deck rack cart with push bar and brake lever; rack frame on a turntable with an upper and a lower spindle (3 notches each, turned-up tips), spring lock pin with quarter-turn holes, turn handle; 1 open parts tray	6 identical flat light metal brackets with a hang hole, each with a fitted wire hook (shank and curl), reused
Service the Trash and Recycling Station	Fixed two-bay bin cabinet braced to the wall, worktop directly above both bays, raised plinth with a taped draw line in front of each bay and a stop at the back of each bay; bay names TRASH (left) and RECYCLING (right); marked cloth home at the front-right corner of the worktop	2 same-size square bins with a front handle and a back-hinged lid with a front lip and lid stop; 1 shallow mixed tray and 2 open stream trays marked TRASH and RECYCLING; 2 liner dispenser boxes; 1 low flat collection cart with braked wheels, bed at bin-rim height; 1 damp cloth	2 full drawstring liners (1 black, 1 clear), light dry waste, max 1.5 kg each, reused; new drawstring liners (black and clear, 1 of each per episode); mixed items , at least 1 per stream, empty and dry (cans, plastic bottles, clean paper or card; wrappers, tissues, paper cups), reused; spots of dried dirt on both lids per episode
5S Reset a Workbench Zone	Fixed workbench with a wall shelf directly above its back strip; bench power strip with an ON/OFF switch and light; zone tag; tool trough fixed along the back strip with 3 labelled open-front slots (driver, pliers, wrench); consumable rack fixed at the right end of the back strip with 2 labelled open-front bays (glove, wipe); checklist bracket bolted at the right end of the front strip; marker clip; 2 marked cloth home spots	Open-top returns bin; screwdriver, pliers, and wrench; 2 damp wipe cloths; marker	3 strays (1 mug, 1 stapler, 1 small parts box), reused; 2 consumable packs (1 glove box, 1 wipe pack), reused; 1 checklist card with at least 1 empty four-box row; a light scatter of loose dust over the working area per episode

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Dress the Bed Pillows and Throw	Made bed with flat tucked sheet and blanket; fixed headboard at the head end; clear floor along the near long side for the base	None beyond the installed bed	2 bare pillows (same size, stacked at the config's pillow spot), 2 pillowcases (same size and colour, folded at the middle of the bed), 1 throw longer than the bed is wide (folded in half at the foot end); all reused
Extract a Part from an Injection Mold Mock	Two half mold mock bolted to a bench, moving half on two guide bars with a front handle; rubber tool mat centred in front of the mold	Flush cutters; pocketed parts tray; scrap cup; reject bin (cutters and output group placed per config)	1 molded plastic part with runner and gate loaded in the cavity per episode; scrap runners for setup test cuts
Insert a Duvet into Its Cover	Bed frame with headboard and mattress wearing a tucked fitted sheet; clear floor along the near long side	None beyond the installed bed	1 duvet (folded in half at the config's duvet spot) and 1 buttoned duvet cover that fits it (hand-gathered into a band at the head end); reused
Operate a Press Brake Mock	Hand lever press brake mock bolted to a bench, lever on the right, die groove, two back stops, set to 90 degrees; blank tray left, finished tray and reject bin right	90 degree angle gauge; rubber gauge mat (placed per config); pocketed finished tray; reject bin	1 flat sheet blank per episode with three numbered bend lines, stacked in the blank tray

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Run a Drill Press Sequence	Powered bench drill press bolted to a bench, green ON button and red OFF paddle on the head, clear chuck guard, feed handle on the right with a set depth stop, sharp 6 mm bit fitted; fixture rail along the back of the drill table with a start stop at its left end; red stop button on the wall with a standby person	Sliding steel fixture plate with back fence, left end stop, wooden backer strip, two lever clamps, and a lock lever; blank tray; tool stand with a swivel-blade deburr tool and a chip brush; chip tray hooked on the drill table; pocketed finished tray (left or right per config)	1 aluminium flat bar blank per episode , 120 x 30 x 5 mm, with three punched and numbered marks 30 mm apart; backer strip replaced when worn
Induct Totes to the Conveyor	Fixed waist-height induct bench, open above; scale platform flush in the middle of the worktop with a painted tote outline; display on the bench front edge (weight, TARE and VERIFY buttons, green and red lights); running takeaway conveyor along the back edge at worktop height with a transfer plate and painted induct line	Open item bin; 2 small nesting totes with a flat label panel on one long side; fixed label dispenser presenting one label at a time with a dry tab	8 identical small boxed items (same weight within a few grams), reused; 2 tote labels per episode
Pick at a Goods-to-Person Port	Fixed port with a waist-height top and a window in the middle, nothing above it; roller lane with a drop stop inside the port that rolls the presented tote out and the next one in on RELEASE; 4 fixed order-slot ledges (S1, S2 left of the window, S3, S4 right), each with a slot light and slot label	Panel on a movable fixed-height stand (display showing PICK n / DONE, lit green CONFIRM and blue RELEASE buttons), linked to the port mock; 4 open order totes fixed in the slots; 3 open source totes	18 small boxed units , 6 of one SKU per source tote, reused; 3 picks loaded into the port mock per episode (pick number 1 to 3, a different slot each)

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Sort a Batch into the Put-Wall	Fixed put-wall of 6 open-front cubbies (A1 to C2, three rows by two columns, each three items wide) with a middle upright; on each cubby lip a cubby label, a lamp with a push button, an order card holder, and a hinged OPEN/FULL flag paddle that clicks up; fixed scanner holster on the middle upright between rows B and C	Auto-read barcode scanner linked to the cubby lamps (put-to-light mock), stays holstered; two-deck batch cart, braked; 1 open batch tote	10 small items with barcode and item label on top, reused; 6 order cards (1, 2, or 3 each, adding to ten), reassigned per episode with the scanner mock
Exchange Totes with an AMR	Fixed waist-height transfer bench, open above, with a one-AMR dock bay in the middle open at the back and a dock stop at the front; painted FULL spot and empty-tote spot on the benches either side (swapped per config); panel on the bay's front edge (DOCKED, FULL TOTE, EMPTY TOTE, DISPATCHED lamps, DISPATCH button, scanner holster)	Wheeled AMR mock with a flat top level with the bench and a four-corner tote nest (drawn out from behind by a helper if passive); handheld scanner linked to the panel; 2 same-size totes with stiff end-wall rims and a front barcode label	A few light loose items filling the full tote about half way, reused
Clear a Conveyor Jam	Short powered belt section at waist height, open above, with low guide rails (lane narrower than a carton is long), reversible flow arrow, jam mark in the middle, and a catch tray at each end; mock HMI on the front frame (STOP, RESET, START buttons; JAM, STOPPED, RUNNING lamps) linked to the belt; log holder and marker clip beside the HMI	Marker	4 identical light cartons labelled A, B, C, D, about twice as long as wide, reused; 1 fault log card per episode (four tick lines)

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Cross-Dock Sort to Four Lanes	Fixed waist-height sort bench, open above; 4 one-parcel mock lane pallets fixed on the top (Lane 1 back left, Lane 2 front left, Lane 3 front right, Lane 4 back right) with lane signs; scanner post at the back middle with a fixed presentation scanner, LANE display, tally card holder, and marker clip	Scanner linked to a lane mock; flat-top infeed cart, braked; marker	10 same-size closed parcels with a barcoded shipping label on one long side, reused, with the parcel-to-lane mapping changed per episode (1 to 4 per lane); 1 tally card per episode (four lines, boxes 0 to 6)
Cycle-Count Six Bins	Fixed three-level shelving (A, B, C) with a shelf above each level and a middle upright; 6 fixed open-front bins (A1 to C2), each one unit wide and six deep, location label under each; work ledge below level C with a front strip open above; count tray with 6 numbered unit-size pockets fixed in the middle of the front strip	Small movable log stand (sheet holder, marker clip, tag cup) placed at the left or right end of the front strip; marker; 3 red COUNT DISCREPANCY tags with spring clips	Small boxed units of 6 SKUs, 2 to 5 per bin plus spares for seeding a one-unit variance in 1 or 2 bins, reused; 1 count sheet per episode (six lines, COUNT and RECOUNT tick rows 0 to 6)
Execute a Slotting Move	Fixed two-level shelving with a shelf above each level and a middle upright; 4 fixed open-front bins (A1, A2, B1, B2), each one unit wide and three deep, with a barcoded location label under each and a clear top-open label holder on each front lip; work ledge below level B with a front strip open above; scanner holster and plan holder on the middle upright	Handheld auto-read barcode scanner linked to the plan (location then card: green light and beep / red light and buzz); 1 small open move tote fixed in the middle of the ledge's front strip	12 small boxed units , 3 each of 4 SKUs, SKU-labelled, reused; 4 stiff bin label cards with SKU name and barcode, reused; 1 slotting plan card per episode naming one level A bin and one level B bin

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Replenish a Pick Face (FIFO)	Fixed shelving with a shelf above the pick-face level and a middle upright; 2 fixed open-front lanes (Lane 1 left, Lane 2 right), each one unit wide and three deep, lane label under each; fixed work ledge below the pick face with a front strip open above (breakdown spot in the middle); log holder and marker clip on the middle upright	Braked flat-top replenishment cart with an open-top flat-carton sleeve on the side facing the base; marker	6 small boxed units of one SKU with large-print date labels (2 older, 4 newer), reused; 1 small corrugated carton with an untaped folded-in bottom and broken-in fold lines (reused until worn); 1 log card per episode (three tick lines)
Reset the Entryway	Bench against a wall under a fixed hook board with 2 coat hooks and 1 bag hook; shallow bowl fixed at the back right; low one shelf shoe rack on the floor at the left end; floor mat between the base and the bench	None beyond the bowl, the rack, and the mat	2 jackets with collar loops , 1 pair of shoes , 1 key set on a ring , 1 bag with a strap ; all reused, staged per config
Seasonal Wardrobe Swap	Bench in front of a wardrobe; fixed short hanging rail above the left end; open shelf above the right end; empty stow bay under the shelf	Lidded storage box with a low divider; 2 hangers	4 soft out-of-season items (stacked per config) and 4 in-season items (2 on hangers, 2 folded flat, in the far half of the box); all reused

Consolidated preparation list

Installed stations

- 1 workbench zone station: fixed bench with a wall shelf above the back strip, power strip with switch and light, zone tag, 3-slot tool trough, 2-bay consumable rack, checklist bracket, marker clip, and 2 marked cloth home spots
- 1 conveyor station
- 1 induct bench: open-above worktop with a flush scale and front-edge display, and a running takeaway conveyor with a transfer plate and induct line along the back

- 1 goods-to-person port: port top with a window over a roller lane with a drop stop, and 4 order-slot ledges with slot lights
- 1 put-wall station: six-cubby put-wall with a lamp, order card holder, and FULL flag on each cubby, and a scanner holster on the middle upright
- 1 AMR transfer station: bench with a middle dock bay and dock stop, FULL and empty-tote spots, and a panel with lamps, DISPATCH, and scanner holster
- 1 jam-clearing conveyor: powered belt section with low guide rails, jam mark, reversible flow arrow, catch trays, and a mock HMI with log holder and marker clip
- 1 cross-dock sort bench: open-above bench with 4 one-parcel lane pallets and signs, and a scanner post with LANE display, tally holder, and marker clip
- 1 cycle-count station: three-level shelving with shelves above, 6 fixed six-deep bins with location labels, a work ledge open above its front strip, and a fixed six-pocket count tray
- 1 slotting station: two-level shelving with shelves above, 4 fixed three-deep bins with location labels and label holders, a work ledge open above its front strip, and a scanner holster and plan holder on the middle upright
- 1 pick-face station: shelving with a shelf above the pick-face level, 2 fixed three-deep lanes, a work ledge open above its front strip, and a log holder with marker clip on the middle upright
- 1 batch-pick walk: two-bay shelving with 8 bins behind a raised track deck with a two-rail track, end blocks, stop lines, and START/PACK signs
- 1 case-pick rack: fixed three-level rack with beams above, 6 labelled case positions, and a scanner holster on the middle upright
- 1 pick station: two fixed shelf bays with shelves above, 8 fixed bins, location labels, and a scanner holster on the middle upright
- 1 put-away shelving station: two-level shelving with shelves above, 4 fixed bins, 2 case lanes with back stops, barcoded location labels, scanner holster, and list box
- 1 spray rack bay: raised deck with a two-rail cart track and load mark, and a fixed three-walled booth mock with a booth stop
- 1 trash and recycling station: fixed two-bay bin cabinet with a worktop above the bays, raised plinth with 2 taped draw lines and 2 bay stops, TRASH and RECYCLING bay names, and a marked cloth home
- 1 dock deck station: smooth-top deck with an empty upper shelf over the lane strip, 2 taped route lanes of 3 numbered spots, 2 route signs, 2 hinged lane flags with catches, and a fixed manifest card holder
- 1 bed station with a cot-size mattress on its frame
- 1 isolated machine with 3 grease zerks
- 1 dishwasher station
- 1 defined hard-floor mopping bay

- 1 sink, kettle, and dispenser station
- 1 pantry shelf-unit station with a shelf above and three zone labels
- 1 refrigerator station
- 1 installed freezer station with a two-level shelf
- 1 microwave station
- 1 fixed mirror and counter station
- 1 3D-printer station
- 1 two-level bookshelf station
- 1 dishwasher unload station: dishwasher, countertop two-shelf cabinet with 2 mug hooks, and cutlery drawer
- 1 dryer unload station: front-loading dryer at arm height, low table in front of it, and hamper stand to its right
- 1 washer load station: front-loading washing machine on a pedestal with a clicking program dial and Start button, water tap shut off, and a low table in front of it
- 1 CNC mock cell: milling mock with two sliding door leaves, clamping vise with a click-out handle, chip trough, front ledge, and a control panel with Cycle Start and DOOR and CYCLE lamps
- 1 bed station with headboard, made bed, pillows, cases, and a throw (pillows and throw)
- 1 injection mold mock bolted to a bench, with centred tool mat
- 1 bed station with headboard and fitted-sheet mattress (duvet)
- 1 press brake mock bolted to a bench, lever on the right, blank tray left, finished tray right
- 1 bench drill press bolted to a bench, powered, with chuck guard, depth stop, fixture rail, and a wall stop button
- 1 entryway bench with hook board (3 hooks), bowl, floor shoe rack, and floor mat
- 1 wardrobe bench with rail above the left end, shelf above the right end, and stow bay
- 1 handheld vacuum station: raised platform with a small bound-edge rug, fixed dock cradle with a spring clip at the right end, and an open-top bin at the left

Reusable tools and staging items

- Outbound lanes: none beyond the installed lanes, flags, and manifest holder
- Induct: item bin, 2 nesting totes, and a label dispenser
- Goods-to-person port: panel on a movable stand, 4 order totes, and 3 source totes
- Put-wall sort: put-to-light scanner mock, braked two-deck batch cart, and 1 batch tote
- AMR exchange: AMR mock with tote nest, handheld scanner, and 2 labelled totes
- Conveyor jam: marker
- Cross-dock sort: lane scanner mock, braked infeed cart, and marker

- Cycle count: log stand with sheet holder, marker clip, and tag cup; marker; 3 discrepancy tags
- Slotting move: handheld auto-read scanner and 1 fixed move tote
- Pick-face replenishment: braked replenishment cart with a flat-carton sleeve, and a marker
- Batch pick: pick cart with end handles, brakes, four-slot tray, and cart scanner
- Case-pick: auto-read scanner and a mock pallet on a braked pallet stand with a ticket clip
- Order pick: auto-read scanner, braked flat-top pick cart with list clip and taped fill and STAGED spots, and 1 order tote with a list pocket
- Put-away: auto-read scanner, braked two-deck put-away cart with list clip, and 1 put tote
- Spray rack: rack cart with push bar and brake, two-spindle notched rack on a locking turntable with turn handle, and 1 parts tray
- Trash and recycling: 2 lidded bins with front handles, 1 mixed tray, 2 stream trays, 2 liner dispenser boxes, 1 braked collection cart, and 1 damp cloth
- Workbench 5S: open-top returns bin, screwdriver, pliers, wrench, 2 damp wipe cloths, and a marker
- Conveyor parts tote, orientation nest, and reject bin
- Fitted sheet: no tools; the bed and mattress are the station
- Lever grease gun, correct coupler and hose, 2 color-coded rags, and grease pencil
- Dishwasher draining board and utensil cup
- Mop, mop pad, bucket with wringer, water jug, and caution sign
- Kettle, fill jug, and absorbent cloth
- Pantry stand and collapsible delivery box
- Refrigerator stock tote
- Freezer work stand, work tray, and 2 labeled bins with front rails
- Heat-proof plate, vented cover, food tub, and spoon
- Spray bottle, squeegee, and lint-free cloth
- 3D-printer build plate, debris tray, plastic scraper, and soft brush
- Bookshelf bin, damp cloth, and solid bookend
- Dishwasher unload: three-compartment cutlery drawer insert
- Dryer unload: hamper with hoop carry handle and lint bin
- Washer load: low hamper and pocket dish
- CNC mock: stock tray, finished tray, and rubber bulb blower in its cup
- Pillows and throw: no tools; the bed, pillows, cases, and throw are the station
- Injection mold: flush cutters, parts tray, scrap cup, and reject bin
- Duvet: no tools; the bed, duvet, and cover are the station
- Press brake: angle gauge, gauge mat, finished tray, and reject bin

- Drill press: sliding fixture plate with two lever clamps and a lock lever, blank tray, tool stand, deburr tool, chip brush, chip tray, and finished tray
- Entryway: bowl, shoe rack, and mat are fixed; no tools
- Wardrobe: lidded divided storage box and 2 hangers
- Handheld vacuum: cordless handheld with latching power button, clear dust cup with flap and catch, and its dock; open-top bin

Episode inputs and consumables

- Workbench 5S: 3 strays (mug, stapler, parts box), 2 consumable packs (glove box, wipe pack), and 1 checklist card with an empty four-box row; strays, packs, tools, and cloths reused, plus a scatter of loose bench dust per episode
- Conveyor: 10 tote parts plus 1 belt misfeed part
- Induct: 8 identical items, reused; 2 tote labels per episode
- Goods-to-person port: 18 SKU-labelled units (6 per source tote), reused; 3 picks loaded into the port mock per episode
- Put-wall sort: 10 barcoded items and 6 order cards, reused, with the item-to-order mapping changed per episode
- AMR exchange: a few light loose items in the full tote, reused
- Conveyor jam: 4 labelled cartons (A to D), reused; 1 fresh fault log card per episode
- Cross-dock sort: 10 labelled parcels, reused, with lane mapping changed per episode; 1 fresh tally card per episode
- Cycle count: units of 6 SKUs (up to 6 per bin) with spares for variance seeding, reused; 1 fresh count sheet per episode
- Slotting move: 12 SKU-labelled units (3 each of 4 SKUs) and 4 bin label cards, reused; 1 slotting plan card per episode
- Pick-face replenishment: 6 date-labelled units (2 older, 4 newer), reused; 1 folding carton, reused until worn; 1 fresh log card per episode
- Batch pick: 24 SKU-labelled units (3 each of 8 SKUs), reused; 1 pick list per episode
- Case-pick: 6 SKU-labelled cartons, reused, plus 1 shuffled pick ticket card per episode
- Order pick: 24 SKU-labelled units (3 each of 8 SKUs), reused, plus 1 shuffled pick list card per episode
- Put-away: 2 SKU-labelled cases and 4 SKU-labelled eaches, reused, plus 1 shuffled put list card per episode
- Spray rack: 6 identical flat metal brackets with fitted wire hooks, reused
- Trash and recycling: 2 full drawstring liners (1 black, 1 clear, max 1.5 kg of light dry waste each) and a mixed set of empty, dry items with at least 1 per stream, all reused; 1 new black and 1 new clear liner and fresh dried dirt on both lids per episode

- Outbound lanes: 6 labelled shoebox-size cases (A1, A2, A3, B1, B2, B3; A3 and B1 weighted with a HEAVY band) and 1 manifest card, all reused
- Fitted sheet: 1 cot-size fitted sheet (reused until its elastic goes slack or a seam opens)
- Greasing: grease for at least 9 strokes and 1 blank lube-card row
- Dishwasher: 12 place-setting items, reused
- Mopping: water to the bucket fill line and 1 clean mop pad
- Sink refill: cold water for the kettle and the dispenser tank, carried by the fill jug
- Pantry: 6 new stock items plus 3 older items for initial placement, and 1 collapsible delivery box (reused until it no longer collapses or stands)
- Refrigerator: 7 new stock items plus 6 older items for initial placement
- Freezer: 6 dated items (3 bags, 3 boxes) and 1 label sheet carrying 2 labels
- Microwave: food for 3 level spoonfuls plus remainder
- Mirror: glass cleaner for at least 6 bursts
- 3D printer: 1 completed print with skirt/purge line and 1 queued job
- Bookshelf: 6 books, 3 non-book objects, and removable shelf dust
- Dishwasher unload: 12 place-setting items (2 plates, 2 bowls, 2 mugs, 6 utensils), reused every episode
- Dryer unload: 6 garments (3 t-shirts, 3 hand towels) reused every episode, and 1 lint sheet per episode
- Washer load: 6 garments (2 t-shirts, 2 hand towels, 2 pairs of pocketed shorts) reused, 1 tissue per episode
- CNC mock: 1 stock block and 1 finished part (reused), about 20 plastic chips re-scattered per episode
- Pillows and throw: 2 bare pillows, 2 cases, and 1 throw (reused)
- Injection mold: 1 molded part with runner per episode, plus scrap runners for setup test cuts
- Duvet: 1 duvet and 1 buttoned cover (reused until a button or seam fails)
- Press brake: 1 numbered sheet blank per episode (consumed by the bends)
- Drill press: 1 marked aluminium blank per episode (consumed by the holes); wooden backer strips as they wear
- Entryway: 2 coats, 1 pair of shoes, 1 key set, 1 bag (reused)
- Wardrobe: 4 out-of-season and 4 in-season soft items, 2 hangers (reused)
- Handheld vacuum: 1 scatter of dry crumbs over the rug per episode; rug and vacuum reused, vacuum kept charged

Points to confirm before procurement

1. **Conveyor part count:** the SOP says the tote contains 10 parts and one part is already on the belt. Confirm whether to stage 11 total parts or take the belt misfeed from the original set

of 10.

2. **Food and cleaning specifications:** the SOPs define functional properties but do not name an exact food, glass-cleaner formula, grease grade, mop-water additive, or book sizes. Select safe, repeatable standard items and record the chosen specifications.
3. **Reusable versus replaced each episode:** define the replacement frequency for rags, cloths, mop pads, lube cards, food, printed parts, simulated refrigerator and pantry stock, pantry delivery boxes, and freezer label sheets.
4. **Lockout control:** the grease task requires a stopped machine with its isolator at OFF and a lockout tag already fitted. Confirm the authorized lockout procedure and responsible person before collection.